

Can a Robot's Skills Be Reusable Building Blocks?

Task add-ons (LoRAs) on a fine-tuned $\pi 0.5$ policy — results, explained simply

FARM · UF850 robot arm · CS153 · 2026-06-02

THE SETUP

We have a robot arm and an AI model that watches two cameras and decides how to move the arm. We taught it four pick-and-place jobs (bottle, bear, duck, hat → box). Then we tested whether each job could be a small, swappable 'skill add-on' on one shared model.

WHAT WE FOUND (one line each)

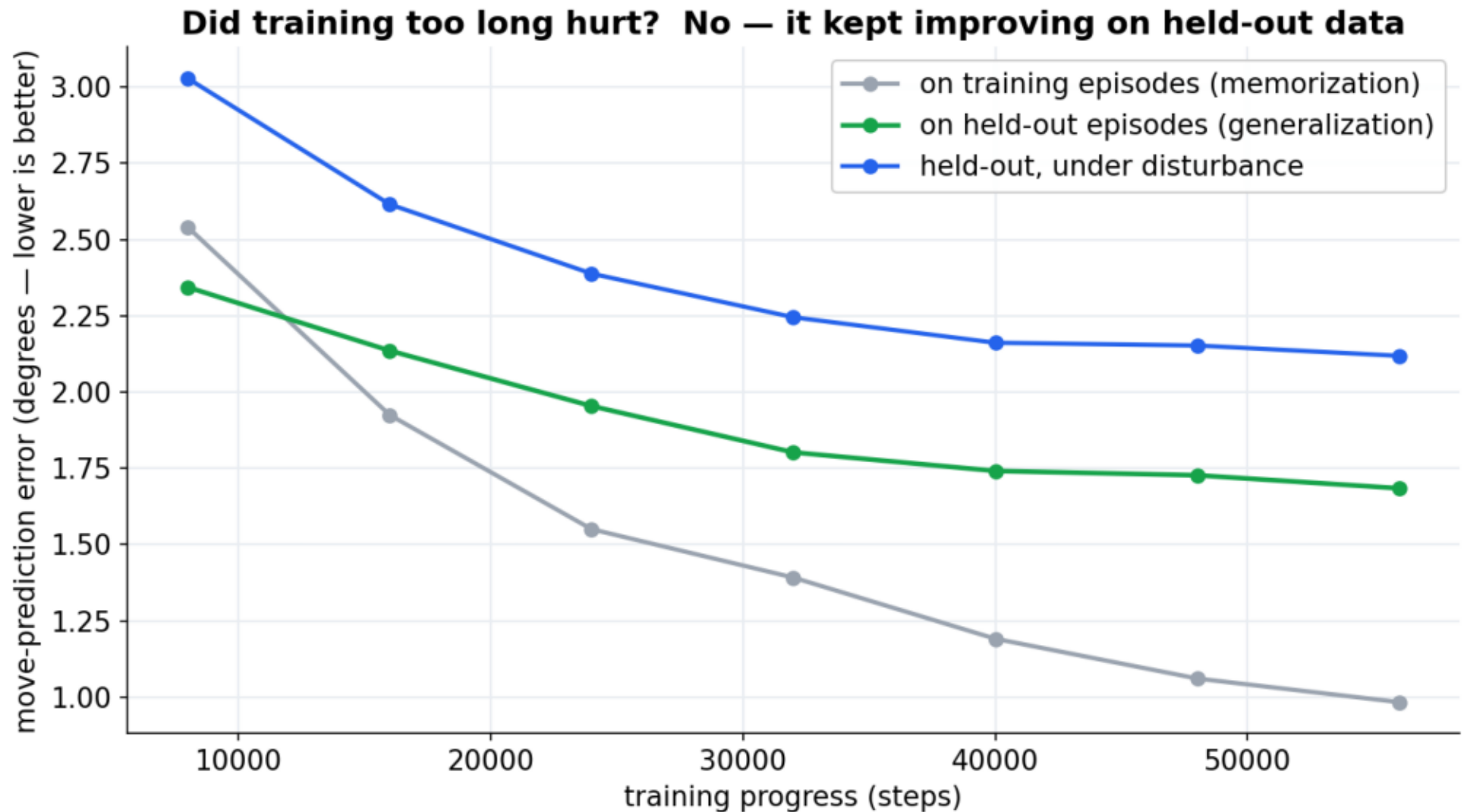
1. The shared model is strong, and training it longer did NOT make it worse.
2. This way of training beats the alternatives (most accurate; handles all 4 objects).
3. Adding a per-task skill add-on makes the robot ~37% more precise – on new episodes.
4. BUT the add-ons are not tidy, comparable 'skill vectors': their raw numbers are dominated by the random starting point of training, not the task. (Train the SAME task twice from different random starts → the two add-ons come out UNRELATED.)

BOTTOM LINE

A skill add-on is a real, useful skill – but you can't compare or combine the add-ons by their raw numbers. A skills library has to compare them by what they DO, or start them all from the same point. The detail that makes this trustworthy: we ran the control that most people skip – and it flipped the conclusion.

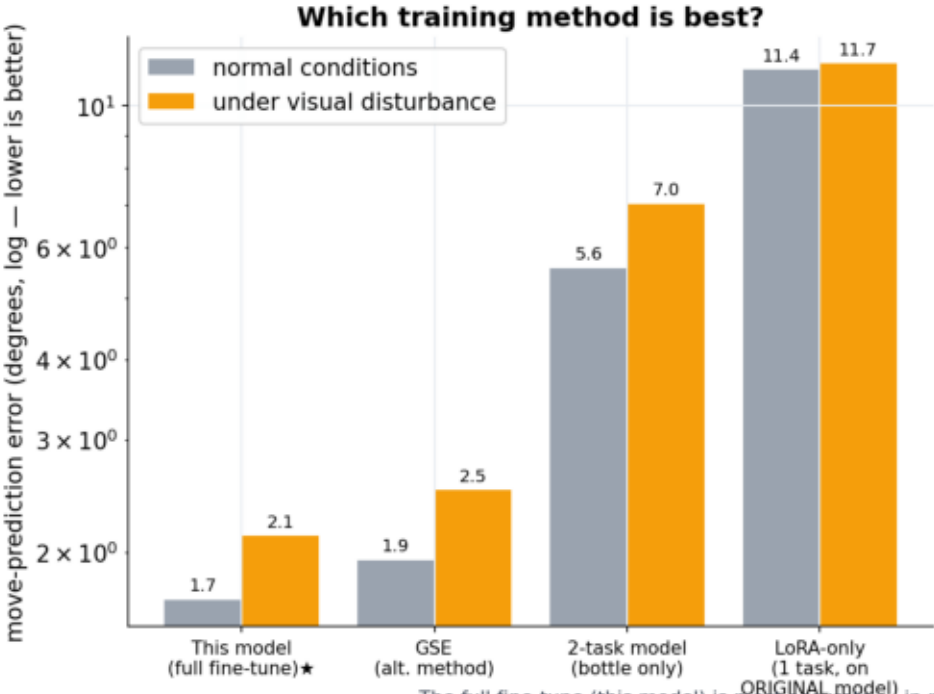
How to read the error numbers: the robot predicts the arm's next angles; we report how far off it is, in degrees (lower = better). 1° is very precise – a clock's minute hand moves 6° per minute. 'Held-out' = recordings the model never saw in training.

Result 1 — The shared model didn't 'over-memorize'

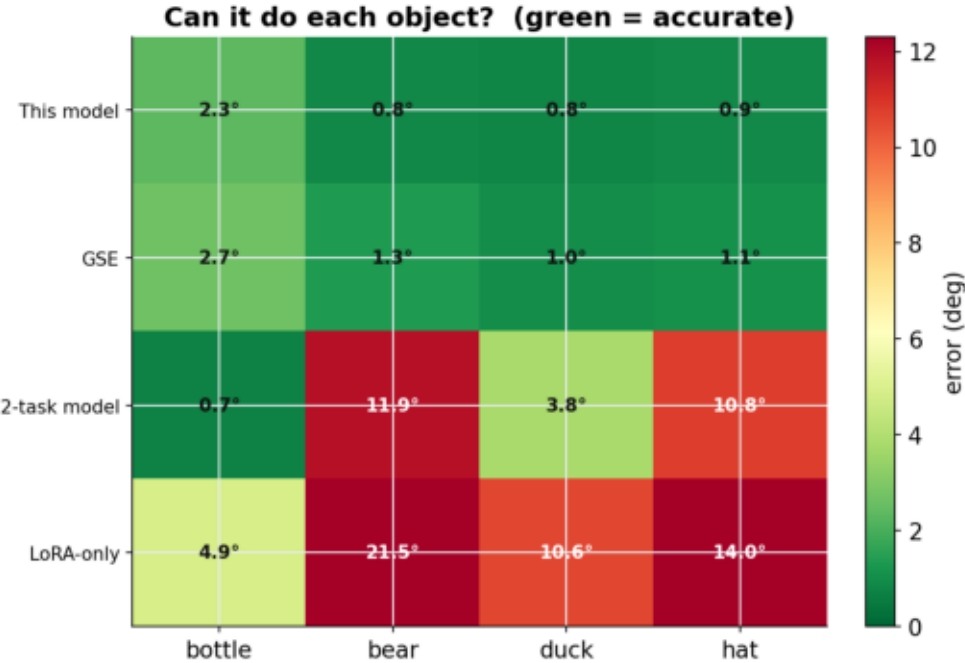


We worried the longest-trained model might 'over-memorize.' It didn't: held-out accuracy improves all the way to the end, so we use the final checkpoint as the base for the skills.

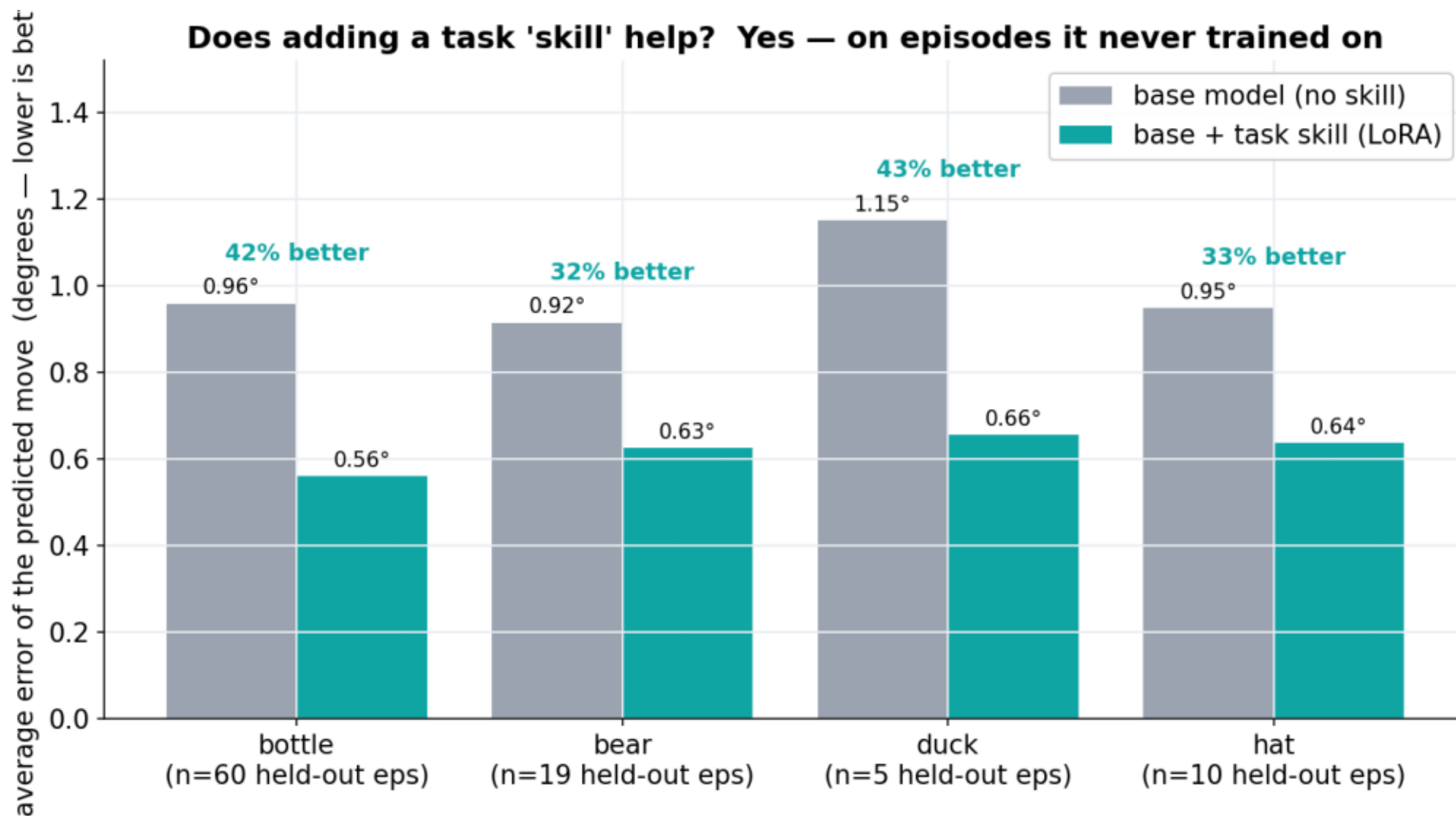
Result 2 — This training method is the most accurate, and covers all 4 objects



The full fine-tune (this model) is most accurate — in normal conditions AND under visual disturbance — and, with GSE, is one of only two trained on all four objects. The two 'bottle-only' methods were trained on bottle alone, so they fail on bear/hat (never seen). NOTE: this 'LoRA-only' is a STANDALONE method on the ORIGINAL model — a weak baseline — NOT the task add-ons in Result 3, which build on THIS strong model and DO help.

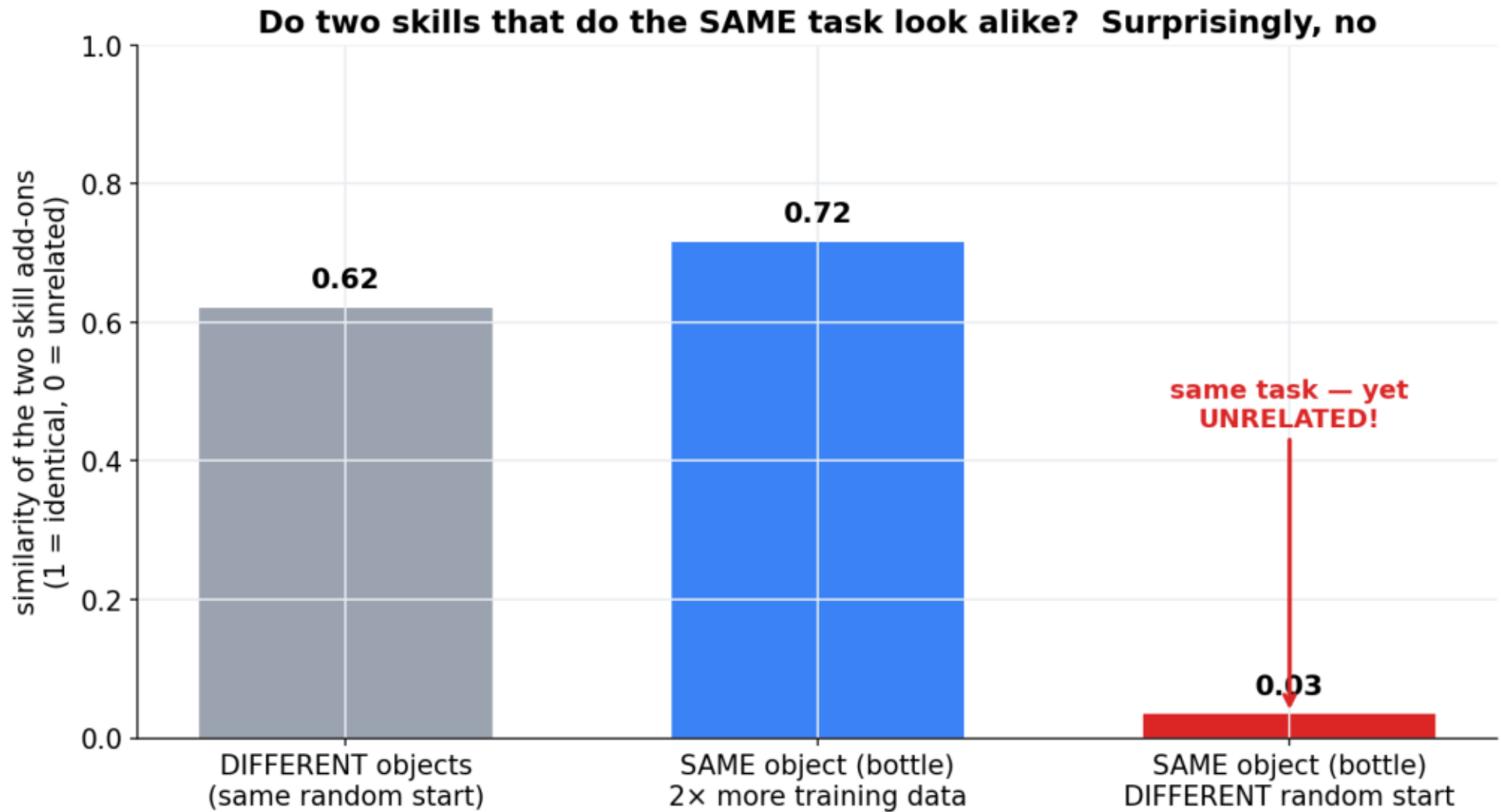


Result 3 — Adding a per-task skill add-on helps (on episodes it never saw)



Each task gets its own small add-on ('skill'). Tested on held-out episodes the add-on never saw, it cuts the robot's move-prediction error by ~30-43%. (1° ≈ very precise; a clock's minute hand moves 6° per minute.)

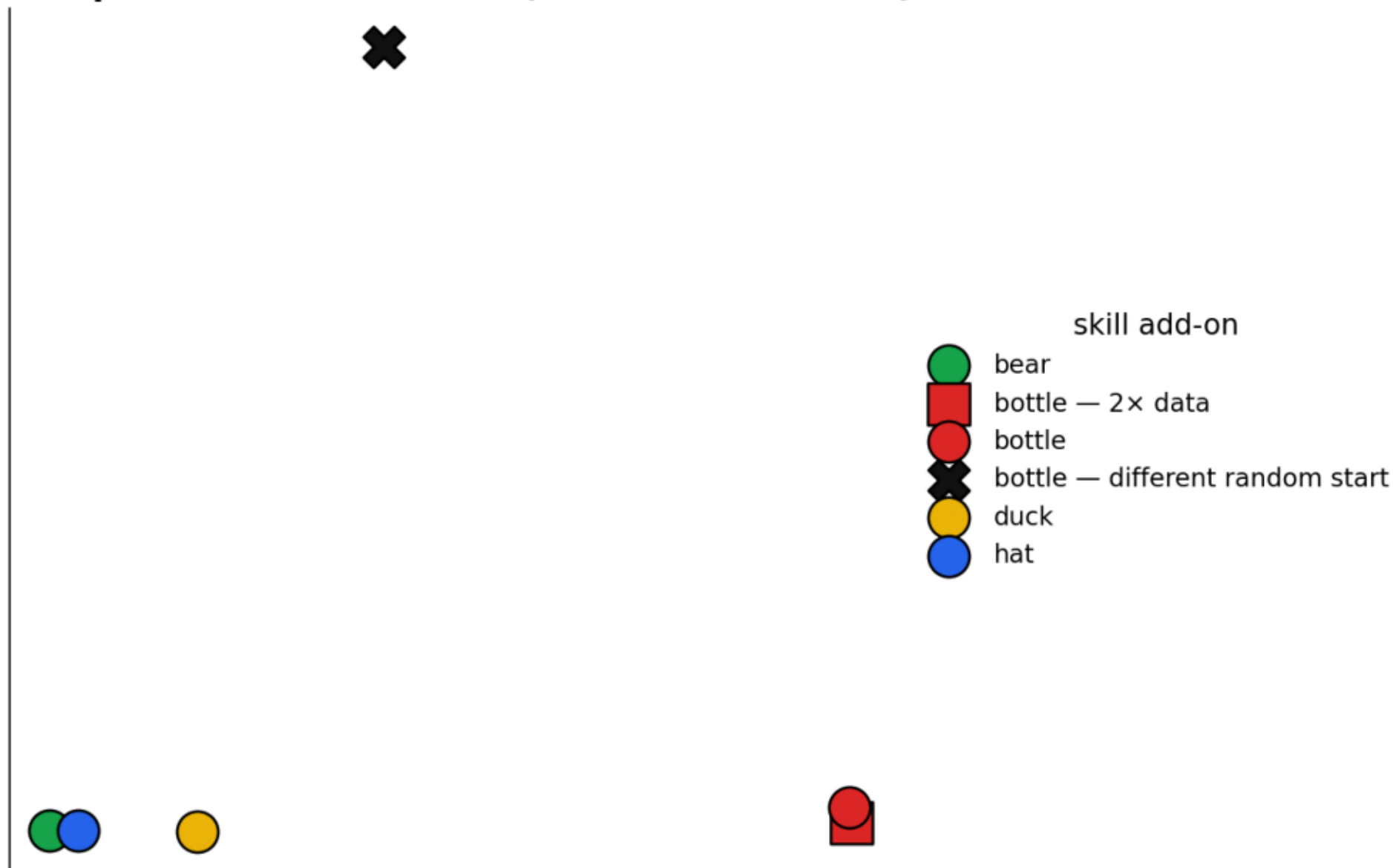
Result 4 — The surprise: same task, different random start ⇒ UNRELATED add-ons



A skill is a set of numbers added to the model. Train the SAME bottle task twice, changing only the random starting point → the two come out essentially UNRELATED (0.03). So the raw numbers mostly reflect the random start, not the task — the task itself adds only a small extra similarity (0.72 vs 0.62).

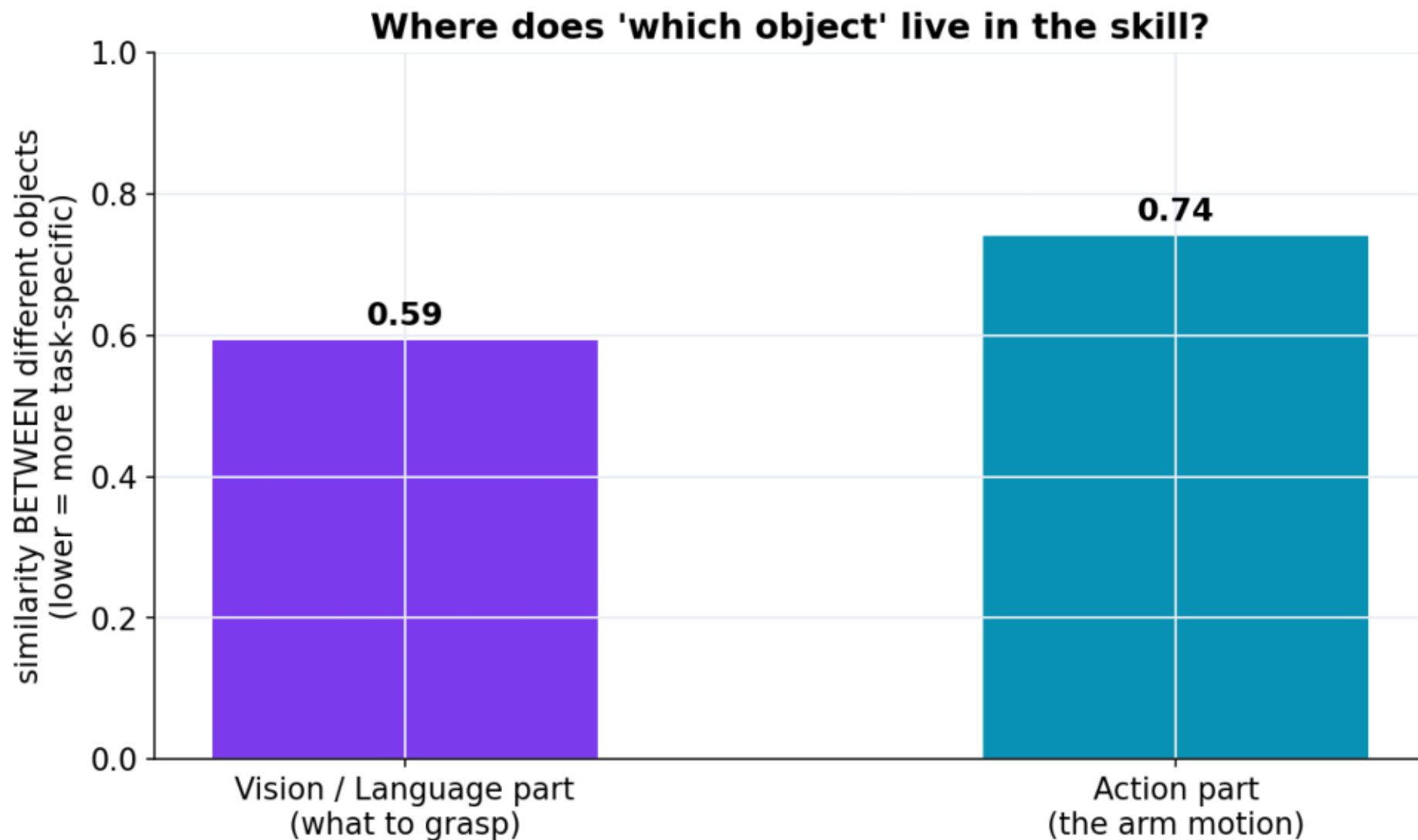
Result 4 — A 'map' of the add-ons: the different-start copy flies off alone

A 'map' of the skill add-ons (closer = more alike)



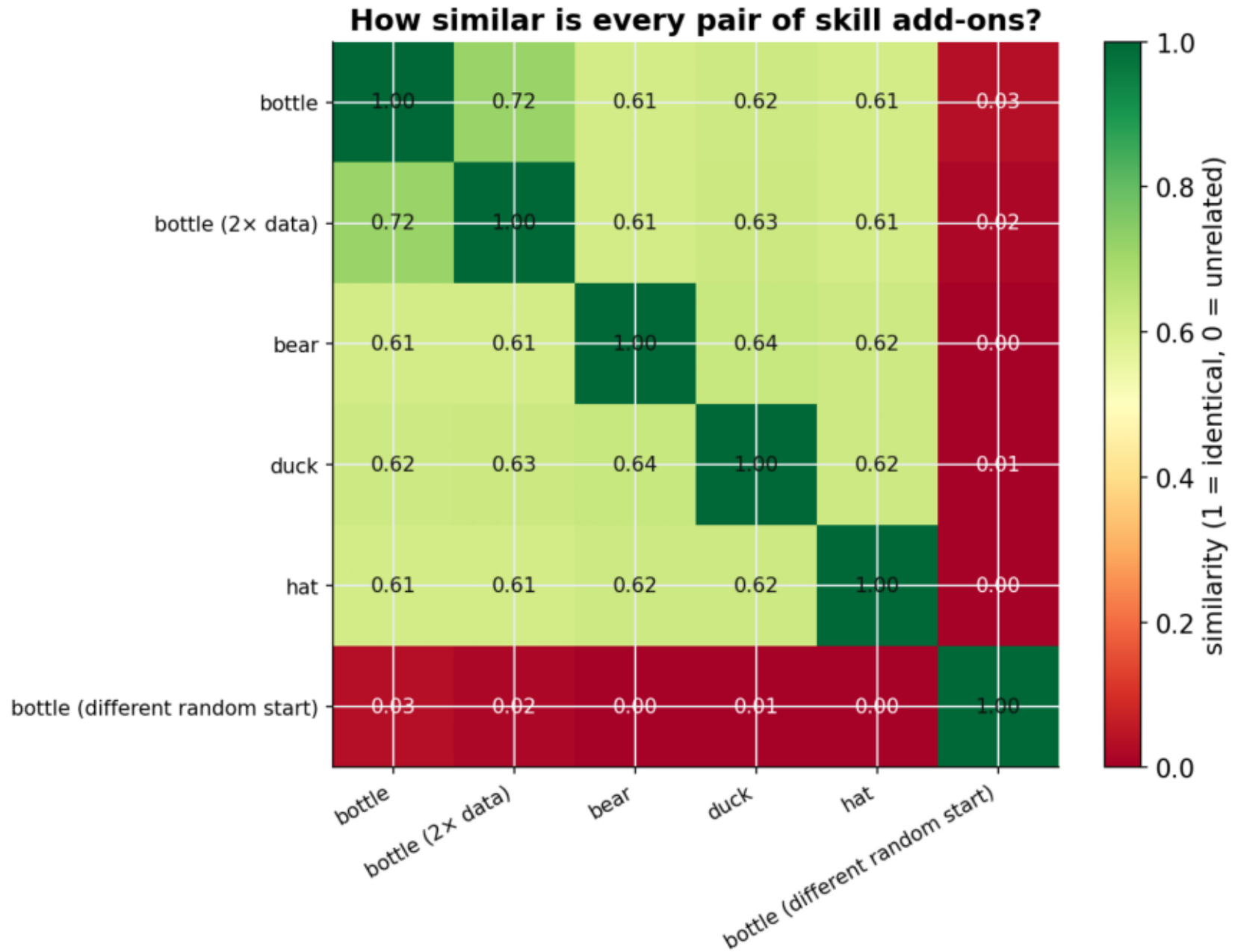
The 'different random start' copy of bottle (black x) sits far off on its own, while the four real tasks barely separate — because the random starting point dominates the raw numbers more than the task does.

Result 4 — Where 'which object' lives: more in the vision/language part



Across different objects, the vision/language adapters differ more (0.59 → more object-specific) while the arm-motion adapters are more shared (0.74). Makes sense: WHAT to grab differs by object; HOW to pick-and-place is similar.

Result 4 — Full similarity table (for reference)



Bright green = alike. The 'different random start' bottle (bottom-right) is unrelated to everything (≈ 0), even to the other bottle skills — the giveaway that the raw numbers track the random start, not the task.

What it all means

The useful, surprising part — and what it implies for building a skills library

FUNCTIONALLY, SKILLS WORK.

- The shared model is strong (Results 1–2) and bolting on a task add-on reliably sharpens it by ~37% on held-out episodes (Result 3).

BUT THE ADD-ON'S RAW NUMBERS ARE NOT A CLEAN 'SKILL COORDINATE' (Result 4).

- They are dominated by the random starting point of training: the SAME task trained twice from different starts gives UNRELATED add-ons (similarity 0.03), while different tasks from the same start look alike (0.62).
- So you cannot reliably compare, average, or do arithmetic on add-ons across runs.
- A faint real task signal does exist once the shared 'starting-point' direction is removed, and it lives more in the model's vision/language part ('what to grab') than in the arm-motion part ('how to move') — which is intuitive.

FOR A REAL SKILLS LIBRARY:

- Either start every skill from the SAME fixed point (so they're comparable), or
- compare/combine skills by what they DO (behavior), not by their raw numbers.

THE NON-OBVIOUS TAKEAWAY:

A LoRA is a genuine functional skill, but NOT a faithful skill vector — and you only find that out if you run the 'same task, different random start' control. Without it, the raw numbers would have fooled us into 'the tasks are similar.'

Honest limitations & glossary

LIMITATIONS

- Error is measured offline (predicted vs demonstrated move), not as success on the physical arm — a strong proxy, but a real robot trial is the gold standard.
- Only 4 objects, so no claims like 'all soft toys cluster.'
- The key 'same task, different start' control was run once (bottle); repeating it per object would make it airtight. Bear/duck/hat held-out sets are small (5–20 eps).

GLOSSARY

- AI model ($\pi_{0.5}$): ~3.3B-number network mapping camera images + arm position → next move.
- Full fine-tune: retrain ALL the model's numbers (powerful, expensive).
- LoRA / skill add-on: a SMALL set of extra numbers added on a frozen model to specialize it for one task — cheap, swappable.
- Error (degrees): how far the predicted joint angles are from the human demo (lower is better; $\sim 1^\circ$ is very precise).
- Held-out: test data never seen in training (honest test of generalization).
- Similarity (0–1): how alike two sets of numbers point (1 identical, 0 unrelated).
- Random start / seed: the random initial values training begins from.
- Domain shift: changing lighting/color/blur to test robustness to a new environment.

Full technical write-up: REPORT.md. All models (base + add-ons) and the dataset are public on HuggingFace; code is in model/ and model/cluster/.